

WJEC CBAC

Predicted Paper

GCE A LEVEL

1520U30-1

ECONOMICS – A2 unit 3

Exploring Economic Behaviour (Predicted Paper – Advanced)

2 hours

Surname: _____ First name(s): _____

Centre Number: _____ Candidate Number: _____

For Examiner's use only

Question	Maximum Mark
Section A 1.	8
2.	7
3.	8
4.	6
5.	11
Section B 6.	10
7.	8
8.	10
9.	12
Total	80

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer all questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
You are reminded of the necessity for good English and orderly presentation in your answers.

SECTION A

Answer all questions in the spaces provided.

1.

Taiwan Semiconductor Manufacturing Company (TSMC) is the world's largest contract chipmaker, producing more than 90% of the world's most advanced semiconductors. TSMC's huge 'gigafabs' cost over US\$20 billion each to build and operate on production runs measured in billions of chips, exploiting substantial internal economies of scale. Its newest 3-nanometre fab in Hsinchu uses extreme ultraviolet lithography machines costing up to US\$380m each; spreading this capital cost over massive output volumes dramatically lowers unit cost.

TSMC's presence has made Taiwan's 'Silicon Shield' region the densest semiconductor cluster in the world. Hundreds of specialist equipment suppliers, materials firms and IP-design houses are co-located nearby. The Taiwanese government has funded the Industrial Technology Research Institute, upgraded the Hsinchu Science Park's transport and power infrastructure, and supported dedicated Masters and PhD programmes at National Taiwan University.

However, TSMC's rapid expansion has created managerial strains. Building new fabs abroad (in Arizona and Kumamoto) has revealed problems transferring the TSMC culture: 30-year engineers have reported burnout, managers struggle to coordinate across global sites, and TSMC's Arizona fab was delayed by 18 months partly due to labour disputes and poor communication between Taiwanese and American staff. Chip yields have occasionally been affected by communication breakdowns inside the firm.

- (a)** Using a diagram and with reference to the data, outline the difference between internal economies of scale and internal diseconomies of scale. **[4]**

Draw your diagram in the space below.

- (b)** Using a diagram, outline how TSMC might benefit from external economies of scale in Hsinchu. **[4]**

Draw your diagram in the space below.

2. The diagram below represents a natural monopoly, such as a national water utility, subject to regulation.

[Diagram: Classic natural monopoly diagram with continuously falling AC and MC curves (LRAC declining over the relevant range). Market demand (AR) curve is downward-sloping. MR is below AR. Three possible regulated price points are typically considered: (i) unregulated profit-maximising output at $MC = MR$; (ii) allocative efficiency at $AR = MC$; (iii) average-cost pricing at $AR = AC$.]

Key curves shown: MC, AC (both falling), AR (demand), MR.

- (a)** On the diagram, indicate the level of output that an unregulated profit-maximising natural monopoly would produce. **[1]**
- (b)** Shade the area representing the supernormal profit earned at the unregulated profit-maximising output. **[2]**
- (c)** Explain why regulators often set a price equal to average cost (AC) rather than equal to marginal cost (MC) for a natural monopoly. **[4]**

3.

YPF (Yacimientos Petrolíferos Fiscales) is Argentina's largest oil and gas producer. Founded as a state enterprise in 1922, YPF was privatised in 1993 and sold to Spain's Repsol, becoming one of Latin America's largest-ever privatisations. In 2012, President Cristina Fernández de Kirchner re-nationalised 51% of YPF, citing underinvestment by Repsol and declining production. Argentina paid US\$5 billion in compensation after prolonged legal disputes.

In 2023–2024, Argentina's new President Javier Milei — a self-described libertarian — announced plans to re-privatise YPF again, along with the national airline Aerolíneas Argentinas, the state media holding, and a number of other state-owned enterprises. Milei argues that state ownership has distorted incentives, accumulated losses (Aerolíneas has required repeated government bailouts), and diverted scarce fiscal resources from public services. Critics argue that previous Argentine privatisations (including YPF itself in 1993) produced uneven results, and that under Milei's 'shock therapy' approach, assets may be sold quickly and below fair value.

Argentina faces particular challenges with privatisation: the peso has lost over 90% of its value since 2018; inflation ran above 200% in 2023; investors face currency controls and risk premiums are high; and Argentina has restructured its sovereign debt nine times since independence. The Vaca Muerta shale formation, in which YPF has major holdings, is one of the world's largest unconventional oil and gas resources — but developing it requires substantial foreign investment.

Evaluate the view that Argentina's planned privatisation programme is likely to be successful. **[8]**

4. In 2023–24, the US Federal Reserve held its federal funds rate at a 23-year high of 5.25–5.50%. Figure 1 shows the impact on a selection of emerging market central bank policy rates.

Figure 1: Central bank policy rates, selected countries, end-2021 to end-2023 (%)

Country	End-2021	End-2022	End-2023
United States (Fed funds)	0.25	4.50	5.50
Argentina	38.00	75.00	133.00
Turkey	14.00	9.00	42.50
Brazil	9.25	13.75	11.75
Mexico	5.50	10.50	11.25
Egypt	8.25	16.25	19.25

Analyse the likely impact of aggressive interest rate rises on aggregate demand in an emerging-market economy such as Argentina. [6]

5. Argentina has suffered chronic inflation for much of its history. In December 2023, monthly inflation reached 25.5% (an annualised rate of over 1500%) following President Milei's initial 54% devaluation of the peso.

Figure 1: Argentina annual CPI inflation rate, 2017–2024 (%)

Year	2017	2018	2019	2020	2021	2022	2023	2024 (est.)
Annual CPI (%)	24	48	54	42	51	95	211	118

Figure 2: Argentina monthly inflation rate by selected categories, December 2023

Category	Food & bev.	Health	Transport	Housing & utilities	Clothing	Education
Monthly inflation rate	29.7%	32.6%	31.7%	13.8%	18.9%	8.7%
Weighting (% of spending)	27	10	14	28	8	13

(a) Using the data in Figure 2, calculate a monthly price index for Argentina in December 2023, taking November 2023 as the base period. You are advised to show your working. [3]

Figure 3: Argentina index of real and nominal wages, 2019–2024 (Dec 2023 = 100)

Year (Dec)	Nominal wage index	Real wage index
2019	8	148
2020	13	140
2021	22	138
2022	45	130
2023	100	100
2024	210	96

Argentine trade unions have historically indexed wages formally to past inflation through 'paritarias' (collective bargaining agreements). In 2024, some leading unions negotiated wage adjustments of 15–25% per month, matching recent monthly inflation.

(b) Assess whether Argentina is likely to experience a continued wage-price spiral in 2024–2025. [8]

SECTION B

Answer all questions in the spaces provided.

Argentina's peso crisis, Milei's 'chainsaw' and the IMF

In December 2023, libertarian economist Javier Milei was inaugurated as Argentina's President after a remarkable electoral victory. Brandishing a chainsaw during campaign rallies to symbolise his plans to cut the state, Milei promised 'shock therapy' to rescue an economy facing collapse. On his first full day in office, Milei's Economy Minister devalued the official peso by 54% and announced sweeping spending cuts.

Figure 1: Argentine peso (ARS) per US dollar, 2019–2024 (official exchange rate)

Date	Jan 2019	Jan 2021	Jan 2023	Dec 2023	Jun 2024	Nov 2024
ARS per US\$	38	85	185	800	900	1,025

(A higher number means a weaker peso — the peso has lost more than 96% of its official US dollar value since 2019.)

Argentina's economic troubles are long-standing but acute. In 2022, public debt stood at 85% of GDP. Around 40% of this debt was denominated in US dollars. By late 2023, 70% of Argentine federal tax revenue was being spent on debt-service costs alone. The Central Bank of Argentina had almost no net foreign reserves — when Milei took office, net reserves were estimated at minus US\$11 billion.

Argentina's parallel exchange rate (the 'blue dollar' rate available on informal markets) diverged wildly from the official rate during periods of capital controls. In mid-2023, the blue dollar rate was more than double the official rate, reflecting severe lack of confidence in the peso.

Argentine firms have been hammered by peso weakness and global supply shocks. Imports priced in dollars — machinery, fuels, fertilisers, pharmaceuticals — have risen sharply in peso terms (Table 1). Argentine manufacturers have laid off workers, reduced output, or exited the market.

Table 1: Prices of selected Argentine imports (US\$ and implied peso terms)

Commodity	2020 US\$ price	2024 US\$ price	% change in peso terms
Pharmaceuticals (index, US\$)	100	128	+2,978%
Diesel fuel (US\$/litre)	0.42	0.88	+4,500%
Natural gas (US\$/mmbtu)	2.60	3.20	+3,000%
Fertiliser (US\$/tonne)	280	380	+3,150%

(Peso-term % changes reflect both US\$ price changes and the peso's depreciation from ~85 to ~1,000 per US\$.)

The sustained depreciation of the peso has caused severe policy challenges. Investors have repeatedly fled the peso, worsening the currency's decline. Between 2021 and 2023, Argentina experienced

capital outflows of over US\$15 billion per year, draining reserves and raising country risk premiums. Sovereign bond yields exceeded 30% at times — making new borrowing effectively impossible.

Figure 2: Argentina Central Bank net international reserves, 2019–2024 (US\$ billion)

Year	2019	2020	2021	2022	2023 (Nov)	2024 (Oct)
Net reserves (US\$bn)	14.0	4.5	2.0	−5.2	−11.0	6.5

To stem the crisis, the Argentine authorities could have continued to raise peso interest rates (they reached 133% by end-2023). However, further increases would have deepened the recession and raised the burden of floating-rate debt. Alternatively, the Central Bank could have intervened in currency markets by selling US dollars — but reserves were negative, making direct intervention almost impossible without new dollar inflows. The third option — which Milei embraced — was orthodox macroeconomic adjustment: a large peso devaluation, deep fiscal cuts, and an IMF agreement to buy time.

Argentina has signed more IMF programmes than any other country — 22 in 60 years, the most recent a US\$44bn Extended Fund Facility programme in 2022, renegotiated in January 2024 to release a further US\$4.7bn tranche and extended in 2024. The IMF described the new programme as providing support for Milei's 'ambitious stabilisation programme' aimed at 'restoring fiscal balance, rebuilding reserves, and lowering inflation'.

Milei's first year in office saw dramatic results. The monthly fiscal deficit was eliminated within months, monthly inflation fell from 25.5% in December 2023 to 2.7% by October 2024, and country risk (EMBI+ spread over US Treasuries) fell from 2,000 to 750 basis points. GDP grew 3.9% quarter-on-quarter in Q3 2024. However, the adjustment was brutal: GDP shrank 3.5% year-on-year in the first half of 2024, real wages in the private registered sector fell by 15% in the first quarter, and the poverty rate rose from 40% to over 55%. Over 300,000 public sector jobs were cut.

Table 2: Selected Argentine development indicators vs upper-middle-income averages

Indicator	Argentina	Upper-middle-income avg	OECD avg
GNI per capita (US\$)	14,230	10,860	42,700
Life expectancy at birth (years)	76.5	75.3	80.3
Adult literacy (%)	99.0	95.5	99.1
Poverty rate (%) (2024)	55.2	—	—
Inequality (Gini)	0.42	0.38	0.316
Access to sanitation (%)	94.5	90.9	99.5
Internet users (%)	87.2	73.4	88.5
Tertiary enrolment (%)	94.3	46.9	71.1

Some economists argue Milei's strategy echoes Argentina's failed 1990s 'Convertibility' regime, which ended in catastrophic collapse in 2001–02. Others argue that this time is different: Milei has cut the fiscal deficit decisively, whereas the 1990s programme relied on privatisation receipts and foreign borrowing; and the productive base (oil, gas, lithium, agriculture) is more diverse today. A possible move to dollarisation (formally replacing the peso with the US dollar) remains on the table but would require massive reserves Argentina does not currently have.

Regardless of the path chosen, the short-term human cost of adjustment has been severe. Food banks report unprecedented demand. Whether Argentina can sustain the reform momentum, rebuild its real economy, and deliver genuine improvements in living standards — rather than just stabilising prices — is the question that will shape South America's second-largest economy for years to come.

6.

(a)

(i) Using the data in Table 1, calculate the percentage change in the US dollar price of diesel fuel between 2020 and 2024. You are advised to show your working. **[2]**

(ii) The peso has depreciated from approximately 85 per US\$ in 2021 to approximately 1,000 per US\$ in 2024. Using this information and your answer to (a)(i), calculate the percentage change in the peso price of diesel fuel between 2020 and 2024, assuming that the peso traded at 65 per US\$ in 2020. You are advised to show your working. **[2]**

(b) With reference to the data and using a cost and revenue diagram, show how the weak peso has resulted in higher prices, fewer sales and falling profits for Argentine firms that import inputs. **[6]**

7. Evaluate the extent to which Argentina's economic difficulties in 2023–2024 were the result of external rather than internal factors. **[8]**

8. Discuss whether the Argentine authorities should have intervened in the ways described in the case to strengthen the peso, rather than pursuing devaluation and fiscal adjustment. **[10]**

9. With reference to the case study, discuss whether living standards in Argentina are likely to be improved by President Milei's 'shock therapy' approach combined with continuing IMF support. **[12]**

END OF PAPER